

# Generation Interconnection System Impact Study Report

## For Queue Project Q15-0516

Granada 138 kV  
February 28, 2017

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0516. The Interconnection Customer proposes a 185.0 MW solar photovoltaic facility interconnecting at the Granada 138 kV substation in the Pennsylvania area. The request was received on September 07, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 247 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$77,783,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                                      | Contingency                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------------------------------|-----------------------------------------------|-------------|--------------|----------|
| GRANADA / BUSINESS PARK 138 KV                                          | BIXBY HILL SUBSTATION / TOMBSTONE 500 KV line | 266.0       | 160          | 70.3     |
| FOURTH BRANCH HYDROELECTRIC FACILITY / ROBBIE 138 KV                    | GRANADA / BUSINESS PARK 138 KV                | 208.8       | 176          | 100.0    |
| HAMILTON / TOMBSTONE SUBSTATION 138 KV L                                | ASH STREET STA 23 / TOMBSTONE 345 KV          | 171.2       | 229          | 28.9     |
| HAMILTON / BAD CREEK 138 KV                                             | ASH ST STATION 223 / TOMBSTONE 345 KV         | 162.1       | 226          | 28.9     |
| STERLINGTON / BUSINESS PARK 138 KV                                      | BUSINESS PARK / TOMBSTONE 138 KV Line         | 160.5       | 170          | 60.1     |
| BUSINESS PARK / TOMBSTONE 138 KV                                        | BUSINESSPARK / STERLINGTON 138 KV             | 150.2       | 207          | 69.0     |
| BAD CRK / PETERSON 138 KV                                               | CONNELL / CHIQUITA 345 KV                     | 147.5       | 220          | 13.0     |
| EDWARDS ST / PETERSON 138 KV Line                                       | CONNELL / CHIQUITA 345 KV                     | 147.3       | 200          | 13.0     |
| STERLINGTON / LA QUINTA 138 KV L                                        | TOMBSTONE / BUSINESS PARK SUBSTATION 138 KV   | 138.6       | 185          | 60.1     |
| EDWARDS ST / PLAINVIEW MUNICIPAL POWER 138 KV                           | CONNELL / CHIQUITA 345 KV                     | 126.7       | 190          | 6.8      |
| FOURTH BRANCH HYDROELECTRIC FACILITY SUBSTATION / JACKSON COUNTY 138 KV | GRANADA / BUSINESS PARK 138 KV line           | 117.9       | 229          | 66.1     |
| ROBBIE / KINDER MORGAN 138 KV L                                         | GRANADA / BUSINESS PARK 138 KV                | 117.2       | 152          | 100.0    |
| LAUDERDALE / ASH ST STATION 223 138 KV                                  | CROMBY / ASH ST STATION 223 345 KV            | 115.1       | 236          | 5.2      |

| Monitored Facility                         | Contingency                                 | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------|---------------------------------------------|-------------|--------------|----------|
| LA QUINTA / TOMBSTONE 138 KV               | BUSINESS PARK / TOMBSTONE 138 KV            | 113.7       | 205          | 60.1     |
| ELK RIV / JACKSON COUNTY 138 KV L          | BUSINESS PARK / GRANADA 138 KV              | 113.0       | 230          | 66.1     |
| HARVEY COUCH / MPC GENERATING 138 KV       | HAMILTON / TOMBSTONE 138 KV                 | 111.2       | 181          | 7.8      |
| OROCOVIS 9902 / LAUDERDALE 138 KV          | CROMBY / ASH STREET STATION 223 345 KV line | 110.6       | 223          | 5.2      |
| SPRECHER SUBSTATION / OROCOVIS 9902 138 KV | CROMBY SUBSTATION / ASH ST STA 223 345 KV   | 105.9       | 210          | 5.2      |
| HALLS / BRADFORD 138 KV                    | ASH ST STA 223 / TOMBSTONE 345 KV           | 102.4       | 165          | 7.5      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| GRANADA / BUSINESS PARK 138 KV                                          | Rebuild line with higher-capacity conductor | \$12,561,000  | 100% (\$12,561,000)   |
| FOURTH BRANCH HYDROELECTRIC FACILITY / ROBBIE 138 KV                    | Rebuild line with higher-capacity conductor | \$12,436,000  | 100% (\$12,436,000)   |
| HAMILTON SUBSTATION / TOMBSTONE SUBSTATION 138 KV L                     | Rebuild line with higher-capacity conductor | \$4,146,000   | 89% (\$3,675,000)     |
| HAMILTON / BAD CREEK 138 KV                                             | Rebuild line with higher-capacity conductor | \$1,509,000   | 89% (\$1,338,000)     |
| BUSINESS PARK / STERLINGTON 138 KV Line                                 | Rebuild line with higher-capacity conductor | \$2,897,000   | 100% (\$2,897,000)    |
| BUSINESS PARK / TOMBSTONE 138 KV                                        | Rebuild line with higher-capacity conductor | \$2,648,000   | 100% (\$2,648,000)    |
| BAD CRK / PETERSON 138 KV                                               | Rebuild line with higher-capacity conductor | \$1,245,000   | 66% (\$816,000)       |
| EDWARDS ST / PETERSON 138 KV Line                                       | Rebuild line with higher-capacity conductor | \$4,159,000   | 66% (\$2,725,000)     |
| STERLINGTON SUBSTATION / LA QUINTA 138 KV                               | Rebuild line with higher-capacity conductor | \$2,496,000   | 100% (\$2,496,000)    |
| EDWARDS ST / PLAINVIEW MUNICIPAL POWER 138 KV                           | Rebuild line with higher-capacity conductor | \$5,845,000   | 39% (\$2,261,000)     |
| FOURTH BRANCH HYDROELECTRIC FACILITY SUBSTATION / JACKSON COUNTY 138 KV | Rebuild line with higher-capacity conductor | \$5,181,000   | 100% (\$5,181,000)    |
| KINDER MORGAN SUBSTATION / ROBBIE 138 KV                                | Rebuild line with higher-capacity conductor | \$5,423,000   | 100% (\$5,423,000)    |
| LAUDERDALE / ASH ST STATION 223 138 KV                                  | Rebuild line with higher-capacity conductor | \$1,129,000   | 29% (\$325,000)       |

| Facility                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------|---------------------------------------------|---------------|-----------------------|
| TOMBSTONE / LA QUINTA 138 KV               | Rebuild line with higher-capacity conductor | \$3,637,000   | 100% (\$3,637,000)    |
| ELK RIV / JACKSON COUNTY 138 KV L          | Rebuild line with higher-capacity conductor | \$1,940,000   | 100% (\$1,940,000)    |
| HARVEY COUCH / MPC GENERATING 138 KV       | Rebuild line with higher-capacity conductor | \$15,442,000  | 100% (\$15,442,000)   |
| OROCOVIS 9902 / LAUDERDALE 138 KV L        | Rebuild line with higher-capacity conductor | \$2,986,000   | 29% (\$859,000)       |
| SPRECHER SUBSTATION / OROCOVIS 9902 138 KV | Reconductor line                            | \$1,057,000   | 29% (\$304,000)       |
| BRADFORD / HALLS 138 KV L                  | Reconductor line                            | \$1,254,000   | 65% (\$819,000)       |
|                                            | Total allocated to project                  |               | \$77,783,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.